



BIO 004 • HUMAN ANATOMY • SOLANO COMMUNITY COLLEGE

Module 3 Lab Structure List.

Muscle • Cardiovascular • Blood • Respiratory

Contents

1. Muscle Tissue and Microanatomy	3
MUSCLE HISTOLOGY	
2. Muscles of the Back and Thorax	5
TRUNK MUSCULATURE	
3. Muscles of the Upper Extremity	6
UPPER LIMB MUSCULATURE	
4. The Heart	7
CARDIOVASCULAR	
5. Blood Vessels, Histology	9
CARDIOVASCULAR	
6. Arteries and Veins of the Thorax and Upper Limb	10
CARDIOVASCULAR	
7. Nerves of the Upper Limb	11
BRACHIAL PLEXUS	
8. Blood	12
BLOOD	

How to use this list

This is a quick reference lab study guide. These are the lab structures and concepts to learn for the lab portion of the exam. Refer to the lab worksheets for the study questions that matter for the course. Treat this as the master list. Occasionally a lab sheet will not contain a structure seen here, so always cross reference against this list. Be prepared to identify structures on models, images, the cadaver, and slides from the microscope.

Some items are tagged with the module they are tested in. You meet the slide in an earlier module and are examined on it here.

Each tissue gets its own block. Tick the box by the name once you can identify it on the slide, write in its function and location, then work down the vertical list and tick each structure you can point to. Anything not in a list is context, not something you will be asked to identify.

READ THIS PART TWICE

This is an extensive list. It would be impossible to test on everything, but everything on this list is testable and you should prepare for it. On occasion I may add something to the list or remove something. Make sure you are in class, and have a person to connect with if you miss class, who can update you on any changes.

CHAPTER 1

Muscle Tissue and Microanatomy

SLIDES AND THE SARCOMERE MODEL

Three tissues on the slide, then the sarcomere on the model. Fill in function and location for each tissue as you go; those columns are what the practical asks you to reason from.

☐ Skeletal muscle, longitudinal section

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|--|-------------------------------------|
| ☐ Muscle fiber | ☐ Peripheral nuclei | ☐ Endomysium |
| ☐ Sarcolemma | ☐ Striations | |
| ☐ Sarcoplasm | ☐ Myofibrils | |

☐ Skeletal muscle, cross section

STRUCTURES TO IDENTIFY

- | | | |
|---|-------------------------------------|--|
| ☐ Muscle fiber, cut across | ☐ Endomysium | ☐ Epimysium |
| ☐ Fascicle | ☐ Perimysium | ☐ Peripheral nuclei |

☐ **Cardiac muscle**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Cardiomyocyte☐ Central nucleus☐ Striations☐ Branching fibers☐ Intercalated discs☐ **Smooth muscle**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Spindle-shaped cell☐ Absence of striations☐ Circular layer☐ Central nucleus☐ Longitudinal layer☐ **The muscle organ model**

STRUCTURES TO IDENTIFY

☐ Epimysium☐ Muscle fiber☐ Belly☐ Perimysium☐ Tendon☐ Origin☐ Endomysium☐ Aponeurosis☐ Insertion☐ Fascicle☐ Deep fascia☐ **The muscle fiber model**

STRUCTURES TO IDENTIFY

☐ Sarcolemma☐ Myofibril☐ Terminal cisternae☐ Sarcoplasm☐ Transverse (T) tubule☐ Triad☐ Myonuclei☐ Sarcoplasmic reticulum☐ Mitochondria☐ **The sarcomere model**

Be prepared to state which filaments are present in each band, line, and zone.

STRUCTURES TO IDENTIFY

☐ Z disc☐ M line☐ Titin☐ A band☐ Zone of overlap☐ Cross bridges☐ I band☐ Thick filament (myosin)☐ H zone☐ Thin filament (actin)

Which regions of the sarcomere narrow when a muscle shortens? _____

Which region never changes length, and why? _____

☐ **Fascicle arrangement, on models**

Name one example muscle for each pattern.

STRUCTURES TO IDENTIFY

☐ Parallel☐ Convergent☐ Multipennate☐ Fusiform☐ Unipennate☐ Circular☐ Bipennate

☐ **Lever systems**

Be prepared to classify a body movement by lever class and to name the fulcrum, effort, and load in it.

STRUCTURES TO IDENTIFY

- | | | |
|----------------------------------|--|---|
| ☐ Fulcrum | ☐ Load | ☐ Second class lever |
| ☐ Effort | ☐ First class lever | ☐ Third class lever |

CHAPTER 2**Muscles of the Back and Thorax****TRUNK MUSCULATURE**

Work these in layers, superficial to deep. For every muscle be prepared to give the origin, the insertion, and the action, not just the name.

☐ **Superficial back****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Trapezius | ☐ Rhomboid major | ☐ Ligamentum nuchae |
| ☐ Latissimus dorsi | ☐ Rhomboid minor | |
| ☐ Levator scapulae | ☐ Thoracolumbar fascia | |

☐ **Erector spinae**

Identify them in order from lateral to medial. Be prepared to name the column from its position alone.

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|--------------------------------------|-----------------------------------|
| ☐ Iliocostalis | ☐ Longissimus | ☐ Spinalis |
|---------------------------------------|--------------------------------------|-----------------------------------|

☐ **Deep back****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|---|
| ☐ Splenius capitis | ☐ Multifidus | ☐ Interspinales |
| ☐ Splenius cervicis | ☐ Quadratus lumborum | ☐ Intertransversarii |
| ☐ Semispinalis | ☐ Rotatores | |

☐ **Thoracic wall**

Be prepared to state the fiber direction of the external and internal intercostals and what each does to the ribs.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ External intercostals | ☐ Transversus thoracis | ☐ Intercostal space |
| ☐ Internal intercostals | ☐ Serratus posterior superior | ☐ Neurovascular bundle in the costal groove |
| ☐ Innermost intercostals | ☐ Serratus posterior inferior | |

☐ **The diaphragm**

Name what passes through each of the three openings.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Central tendon | ☐ Costal part | ☐ Esophageal hiatus (T10) |
| ☐ Right crus | ☐ Sternal part | ☐ Aortic hiatus (T12) |
| ☐ Left crus | ☐ Caval opening (T8) | ☐ Phrenic nerve |

☐ **Anterior thorax****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|--|
| ☐ Pectoralis major | ☐ Serratus anterior | ☐ External oblique, upper slips |
| ☐ Pectoralis minor | ☐ Subclavius | ☐ Clavipectoral fascia |

Which nerve supplies serratus anterior, and what deficit follows its injury?

CHAPTER 3**Muscles of the Upper Extremity****SHOULDER, ARM, FOREARM, HAND**

Learn these by compartment. On a cadaver or a model the compartment tells you the action and the nerve before you have identified the individual muscle.

☐ **Rotator cuff**

Name the tubercle each one reaches and the movement each one produces.

STRUCTURES TO IDENTIFY

- | | |
|--|--|
| ☐ Supraspinatus | ☐ Teres minor |
| ☐ Infraspinatus | ☐ Subscapularis |

☐ **Scapulohumeral, other****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Deltoid, anterior fibers | ☐ Deltoid, posterior fibers | ☐ Coracobrachialis |
| ☐ Deltoid, middle fibers | ☐ Teres major | |

☐ **Arm, anterior compartment****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Biceps brachii, long head | ☐ Bicipital aponeurosis | ☐ Coracobrachialis |
| ☐ Biceps brachii, short head | ☐ Brachialis | |

☐ **Arm, posterior compartment****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Triceps brachii, long head | ☐ Triceps brachii, medial head | ☐ Radial groove |
| ☐ Triceps brachii, lateral head | ☐ Anconeus | |

☐ **Forearm, anterior compartment****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Brachioradialis | ☐ Flexor carpi ulnaris | ☐ Pronator quadratus |
| ☐ Pronator teres | ☐ Flexor digitorum superficialis | ☐ Medial epicondyle, common flexor origin |
| ☐ Flexor carpi radialis | ☐ Flexor digitorum profundus | |
| ☐ Palmaris longus | ☐ Flexor pollicis longus | |

☐ **Forearm, posterior compartment**

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Extensor carpi radialis longus | ☐ Extensor carpi ulnaris | ☐ Extensor pollicis longus |
| ☐ Extensor carpi radialis brevis | ☐ Supinator | ☐ Extensor indicis |
| ☐ Extensor digitorum | ☐ Abductor pollicis longus | ☐ Lateral epicondyle, common extensor origin |
| ☐ Extensor digiti minimi | ☐ Extensor pollicis brevis | |

☐ **Hand, intrinsic muscles**

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Thenar eminence | ☐ Adductor pollicis | ☐ Opponens digiti minimi |
| ☐ Abductor pollicis brevis | ☐ Hypothenar eminence | ☐ Lumbricals |
| ☐ Flexor pollicis brevis | ☐ Abductor digiti minimi | ☐ Palmar interossei |
| ☐ Opponens pollicis | ☐ Flexor digiti minimi brevis | ☐ Dorsal interossei |

☐ **Landmarks and retinacula**

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Cubital fossa | ☐ Extensor retinaculum | ☐ Extensor expansion |
| ☐ Carpal tunnel | ☐ Palmar aponeurosis | |
| ☐ Flexor retinaculum | ☐ Anatomical snuffbox | |

Name the two common origins that most superficial forearm muscles share.

CHAPTER 4

The Heart

MODEL, CADAVER, AND SLIDE

Work the heart from the outside in: sac, wall, chambers, valves, then the vessels on its surface. Be prepared to trace blood through it in order on any specimen you are given.

☐ **Position and coverings**

STRUCTURES TO IDENTIFY

- | | | |
|--------------------------------------|---|---|
| ☐ Mediastinum | ☐ Anterior (sternocostal) surface | ☐ Parietal serous pericardium |
| ☐ Apex | ☐ Inferior (diaphragmatic) surface | ☐ Visceral serous pericardium (epicardium) |
| ☐ Base | ☐ Fibrous pericardium | ☐ Pericardial cavity |

☐ **The heart wall**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|-------------------------------------|-------------------------------------|--------------------------------------|
| ☐ Epicardium | ☐ Myocardium | ☐ Endocardium |
|-------------------------------------|-------------------------------------|--------------------------------------|

☐ **External surface features****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|--|
| ☐ Coronary sulcus | ☐ Posterior interventricular sulcus | ☐ Left auricle |
| ☐ Anterior interventricular sulcus | ☐ Right auricle | ☐ Ligamentum arteriosum |

☐ **Chambers and internal features****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Right atrium | ☐ Fossa ovalis | ☐ Chordae tendineae |
| ☐ Right ventricle | ☐ Pectinate muscles | ☐ Moderator band |
| ☐ Left atrium | ☐ Crista terminalis | ☐ Conus arteriosus |
| ☐ Left ventricle | ☐ Sinus venarum | ☐ Aortic vestibule |
| ☐ Interatrial septum | ☐ Trabeculae carneae | |
| ☐ Interventricular septum | ☐ Papillary muscles | |

☐ **Valves**

For each valve, name what it separates and what it prevents.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Tricuspid valve | ☐ Pulmonary semilunar valve | ☐ Fibrous skeleton |
| ☐ Mitral (bicuspid) valve | ☐ Aortic semilunar valve | |

☐ **Great vessels****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Superior vena cava | ☐ Right pulmonary artery | ☐ Ascending aorta |
| ☐ Inferior vena cava | ☐ Left pulmonary artery | ☐ Aortic arch |
| ☐ Pulmonary trunk | ☐ Pulmonary veins | ☐ Descending thoracic aorta |

☐ **Coronary circulation****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Right coronary artery | ☐ Left coronary artery | ☐ Coronary sinus |
| ☐ Right marginal artery | ☐ Anterior interventricular artery (LAD) | ☐ Great cardiac vein |
| ☐ Posterior interventricular artery (PDA) | ☐ Circumflex artery | ☐ Middle cardiac vein |
| | ☐ Left marginal artery | ☐ Small cardiac vein |

☐ **Conduction system, on the model****STRUCTURES TO IDENTIFY**

- | | | |
|----------------------------------|--|---|
| ☐ SA node | ☐ AV bundle (bundle of His) | ☐ Left bundle branch |
| ☐ AV node | ☐ Right bundle branch | ☐ Purkinje fibers |

☐ **Cardiac muscle slide**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|-------------------------------------|
| ☐ Cardiomyocyte | ☐ Branching fibers | ☐ Striations |
| ☐ Central nucleus | ☐ Intercalated discs | |

Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.

CHAPTER 5

Blood Vessels, Histology

SLIDES AND WALL MODELS

Almost every question here is answered from the wall. Identify the tunics first, then use their proportions to name the vessel.

☐ Artery, cross section

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---------------------------------------|
| ☐ Tunica intima | ☐ Tunica media | ☐ Vasa vasorum |
| ☐ Endothelium | ☐ Smooth muscle | ☐ Lumen |
| ☐ Basement membrane | ☐ External elastic lamina | |
| ☐ Internal elastic lamina | ☐ Tunica externa | |

☐ Vein, cross section

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--------------------------------|
| ☐ Tunica intima | ☐ Tunica externa | ☐ Lumen |
| ☐ Tunica media | ☐ Venous valve | |

Name the two features you would use first to tell an artery from a vein on a slide.

☐ Elastic artery

Name two elastic arteries you would expect to see labelled.

STRUCTURES TO IDENTIFY

- | | |
|---|--|
| ☐ Elastic lamellae | ☐ Internal elastic lamina |
| ☐ Thick tunica media | ☐ External elastic lamina |

☐ Muscular artery

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Prominent internal elastic lamina | ☐ Smooth muscle of the tunica media | ☐ Thin external elastic lamina |
|--|--|---|

☐ Arteriole and capillary bed

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Arteriole | ☐ Capillary | ☐ Postcapillary venule |
| ☐ Metarteriole | ☐ Capillary bed | ☐ Muscular venule |
| ☐ Precapillary sphincter | ☐ Thoroughfare channel | |

☐ **Capillary types**

For each type, name one organ where it is found and state what can cross its wall.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Continuous capillary | ☐ Sinusoid | ☐ Fenestration |
| ☐ Fenestrated capillary | ☐ Intercellular cleft | ☐ Basement membrane |

☐ **Fetal circulation, on the model**

Name the adult remnant of each fetal structure.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Placenta | ☐ Ductus venosus | ☐ Ductus arteriosus |
| ☐ Umbilical vein | ☐ Foramen ovale | ☐ Umbilical arteries |

CHAPTER 6**Arteries and Veins of the Thorax and Upper Limb****NAMED VESSELS, CADAVER AND MODEL**

These are routes. Trace each one continuously with your finger before you try to name segments, and learn the landmark where each name changes.

☐ **Aortic arch and its branches**

Be prepared to explain why the right and left sides are not symmetrical.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Ascending aorta | ☐ Right common carotid artery | ☐ Left subclavian artery |
| ☐ Aortic arch | ☐ Right subclavian artery | ☐ Descending thoracic aorta |
| ☐ Brachiocephalic trunk | ☐ Left common carotid artery | |

☐ **Other thoracic arteries****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|--|
| ☐ Internal thoracic artery | ☐ Bronchial arteries | ☐ Thyrocervical trunk |
| ☐ Anterior intercostal arteries | ☐ Esophageal arteries | ☐ Costocervical trunk |
| ☐ Posterior intercostal arteries | ☐ Vertebral artery | |

☐ **Arteries of the upper limb**

Name the landmark at which the subclavian becomes the axillary, and the axillary becomes the brachial.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Subclavian artery | ☐ Anterior circumflex humeral artery | ☐ Ulnar artery |
| ☐ Axillary artery | ☐ Posterior circumflex humeral artery | ☐ Common interosseous artery |
| ☐ Thoracoacromial artery | ☐ Brachial artery | ☐ Superficial palmar arch |
| ☐ Lateral thoracic artery | ☐ Deep brachial artery | ☐ Deep palmar arch |
| ☐ Subscapular artery | ☐ Radial artery | ☐ Digital arteries |

☐ **Superficial veins of the upper limb****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Cephalic vein | ☐ Median cubital vein | ☐ Dorsal venous network of the hand |
| ☐ Basilic vein | ☐ Median antebrachial vein | |

☐ **Deep veins of the thorax and upper limb**
STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Radial veins | ☐ Subclavian vein | ☐ Azygos vein |
| ☐ Ulnar veins | ☐ Internal jugular vein | ☐ Hemiazygos vein |
| ☐ Brachial veins | ☐ Brachiocephalic veins | ☐ Internal thoracic vein |
| ☐ Axillary vein | ☐ Superior vena cava | |

Why is there one brachiocephalic artery but two brachiocephalic veins?

CHAPTER 7**Nerves of the Upper Limb****THE BRACHIAL PLEXUS**

Build the plexus by level before you learn individual nerves. The counts, five roots, three trunks, six divisions, three cords, five branches, are how you check your own work.

☐ **Levels of the plexus**

The cords are named for their position around the axillary artery. Identify the artery first.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Roots, C5 to T1 | ☐ Lower trunk | ☐ Lateral cord |
| ☐ Upper trunk | ☐ Anterior divisions | ☐ Posterior cord |
| ☐ Middle trunk | ☐ Posterior divisions | ☐ Medial cord |

☐ **Terminal branches**

For each nerve, name the compartment it supplies and the deficit its injury produces.

STRUCTURES TO IDENTIFY

- | | | |
|---|---------------------------------------|---|
| ☐ Musculocutaneous nerve | ☐ Ulnar nerve | ☐ Axillary nerve |
| ☐ Median nerve | ☐ Radial nerve | |

☐ **Branches above the terminal nerves**
STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Dorsal scapular nerve | ☐ Lateral pectoral nerve | ☐ Lower subscapular nerve |
| ☐ Long thoracic nerve | ☐ Medial pectoral nerve | ☐ Medial cutaneous nerve of the arm |
| ☐ Suprascapular nerve | ☐ Thoracodorsal nerve | ☐ Medial cutaneous nerve of the forearm |
| ☐ Nerve to subclavius | ☐ Upper subscapular nerve | |

☐ **Nerves at their vulnerable points**

Name the injury associated with each site and the sign it produces.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Axillary nerve at the surgical neck of the humerus | ☐ Ulnar nerve behind the medial epicondyle | ☐ Long thoracic nerve on the chest wall |
| ☐ Radial nerve in the radial groove | ☐ Median nerve in the carpal tunnel | |

☐ **Thoracic nerves****STRUCTURES TO IDENTIFY**

- | | |
|---|--|
| ☐ Intercostal nerves | ☐ Vagus nerve in the thorax |
| ☐ Phrenic nerve | ☐ Sympathetic trunk |

CHAPTER 8**Blood****THE SMEAR**

One slide, and every question comes off the nucleus. Identify each cell by the shape of its nucleus first and the granules second.

☐ **Whole blood, in a spun tube****STRUCTURES TO IDENTIFY**

- | | |
|---------------------------------------|--|
| ☐ Plasma layer | ☐ Erythrocyte layer |
| ☐ Buffy coat | ☐ Hematocrit |

☐ **Erythrocytes**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Biconcave disc | ☐ Absence of a nucleus |
| ☐ Pale central area | ☐ Rouleaux formation |

☐ **Granulocytes****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Neutrophil | ☐ Eosinophil | ☐ Basophil |
| ☐ Multi-lobed nucleus | ☐ Bilobed nucleus | ☐ Nucleus obscured by dark granules |
| ☐ Fine pale granules | ☐ Coarse red-orange granules | |

☐ **Agranulocytes****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|--|
| ☐ Lymphocyte | ☐ Monocyte | ☐ Abundant pale cytoplasm |
| ☐ Large round nucleus with thin rim of cytoplasm | ☐ Kidney-shaped nucleus | |

☐ **Platelets and formation****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---------------------------------------|
| ☐ Platelet (thrombocyte) | ☐ Red bone marrow | ☐ Reticulocyte |
| ☐ Megakaryocyte | ☐ Hemocytoblast | |

Order the five leukocytes from most to least numerous.
